

Inshaf Rajaei

AI / Machine Learning Engineer | Computer Vision | Data Science

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EDUCATION

University of Ruhuna

Sri Lanka

BSc Honours in Engineering (Computer Engineering) | **CGPA: 3.455**

Expected 2027

- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Computer Vision, Probability & Statistics.

TECHNICAL SKILLS

Programming & Web: Python, JavaScript, React, C++, C

Machine Learning: Scikit-learn, XGBoost, SVM, Decision Trees, KNN, Hyperparameter Tuning

Deep Learning & Vision: TensorFlow, Keras, Neural Networks, OpenCV, YOLOv8, DeepSORT

Data Science: Pandas, NumPy, Exploratory Data Analysis (EDA), Data Visualization, Cross-Validation

Deployment / MLOps: FastAPI, Streamlit, Docker, Git/GitHub

ENGINEERING PROJECTS

Fraud Detection System (Epic Lanka) | Python, XGBoost, FastAPI, React | [GitHub](#)

- Developed a Fraud Detection Engine using XGBoost on a synthetic credit card data set with 90% normal and 10% fraud patterns.
- Engineered 30 features based on the ISO 8583 standard and utilized hyperparameter tuning to optimize evaluation metrics.
- Developed a FastAPI microservice to deploy the Fraud Detection Engine in a production environment with end-to-end scoring latency between 5 and 20 ms.
- Built a React frontend dashboard to display real-time transaction scores and let administrators easily monitor the fraud alerts.
- Configured automatic approval, escalation, or rejection of transactions based on risk thresholds stored in a database with minimal human intervention required.

Vehicle Speed Detection System | Python, YOLOv8, DeepSORT, OpenCV

- Developed a live vehicle tracking pipeline that leverages YOLOv8 neural networks and the Deep SORT algorithm.
- Determined the physical speeds of moving vehicles via utilization of a homography matrix that transfers images into a flat, bird's eye view perspective.
- Processed live traffic feeds in excess of 30 FPS via utilization of standard hardware configurations.
- Tested the developed pipeline against 4,000+ frames to ensure accuracy in spatial calculations and tracking.

Automated Waste Classification System | Python, TensorFlow, Keras, CNNs

- Created an automated image classification system that utilized Convolutional Neural Networks (CNNs) to classify waste into 6 distinct classes.
- Applied transfer learning, leveraging the lightweight MobileNetV2 model and optimizing it for real-time deployment on the edge in resource-constrained environments.
- Applied data augmentation methods to balance the TrashNet dataset and configured the model with an Adam optimizer to achieve an accuracy target above 90%.

Loan Approval Prediction Model | Python, Scikit-learn, FastAPI, Streamlit | [GitHub](#)

- Trained an SVM and Decision Tree classifier on 4,269 financial records to classify whether a borrower's loan is approved or not.
- Cleaned data through careful data scaling, cross-validation, and data exploration.
- Achieved 97.78% accuracy with Decision Tree classifier over SVM's 91.69% and developed an interactive data visualization dashboard using Streamlit.

ACHIEVEMENTS

2nd Runner-Up – Code Night 2025 Hackathon | Certix (Private) Limited

Sept. 2025

- Demonstrated rapid software development and innovation in a competitive, time-constrained environment organized by DEIE.

REFERENCES

Dr. Prabath Weerasinghe | Senior Lecturer, University of Ruhuna | weera@eie.ruh.ac.lk | +94 71 705 6638